ST512-SP26-Homework-1

Jason Menjivar-Hernandez

Problem 1

n = 36

$\bar{x}$ = 23.4

sd (s) = 7.2

CI: $\bar{x}\pm$ t* x SE

SE = $\frac{s}{\sqrt{n}}$

SE = $\frac{7.2}{\sqrt{36}}$ = 1.2

t* = qt(1-(.04/2), 36) = 2.13

CI: 23.4 $\pm$ 2.13 * 1.2

96% CI: (20.84,25.96)

Problem 2

$H_o : \mu = 3.5$

$H_A : \mu < 3.5$

Will be conducting a left-tailed test because we want to know if the drug delivers relief faster (i.e., less than 3.5 minutes).

n = 50

$\bar{x}$ = 3.1

s = 1.5

α = 0.05

SE = $\frac{s}{\sqrt{n}} = \frac{1.5}{\sqrt{50}}$ = 0.212

t-stat = $\frac{\bar{x}-\mu_o}{SE}$ = $\frac{-0.4}{0.212}$ = -1.887

p-value = pt(t.value, df) = 0.033

p-value < α, we reject H_o

Therefore, there is sufficient evidence at the 5% confidence interval to conclude that the newly developed pain reliever delivers perceptible pain relief more quickly than the standard pain reliever.